

Module 5

Input and Output

Computer Science Department · School of ICT · Copperbelt University

Dr Z. Lifelo

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Module 5 — Lecture Overview

Six sections — from I/O fundamentals through manual input, digital input, physical output, digital output, to accessibility and device management:

A

Input and Output Fundamentals

Data definition · Types of input · Commands & user responses · Four types of output: text, graphics, audio, video

B

Manual Input Devices

Keyboards · Mouse & pointing devices · Touch screens & gestures (Table 6-1) · Pen input · Stylus · Graphics tablet

C

Digital Input Devices

Voice & audio input · Video input · Motion/gesture recognition · NFC · Scanners · OCR/OMR · Barcodes & QR · RFID · MICR

D

Physical Output — Printers

Hard copy vs soft copy · Portrait vs landscape · Non-impact printers (inkjet, laser, photo, thermal, 3D, mobile, plotter) · Impact printers

E

Digital Output — Displays, Audio & Projectors

LCD · LED · OLED · AMOLED · Display quality metrics · GPU · Speakers & headphones · Data projectors · Interactive whiteboards

F

Accessibility, Customisation & E-Waste

Adaptive input/output devices · Braille displays · OCR readers · Device customisation · E-waste radiation risks & strategies

Input and Output Fundamentals

Data · Commands · User responses · Four types of output

How Do You Use Input?

Data: A collection of unprocessed items, including text, numbers, images, audio, and video. A variety of options are used to input data and instructions into a computer. (Campbell et al., 2023)

Data Input: Raw unprocessed items entered into a computer: text, numbers, images, audio, and video. Data becomes information after the computer processes it. A variety of options are used to input data and instructions into a computer.

Command: An instruction that causes a program or app to perform a specific action. Programs and apps respond to commands that a user issues. (Campbell et al., 2023)

User Response: An instruction a user issues by responding to a message displayed by a program or app. A response to the message instructs the program or app to perform certain actions. (Campbell et al., 2023)

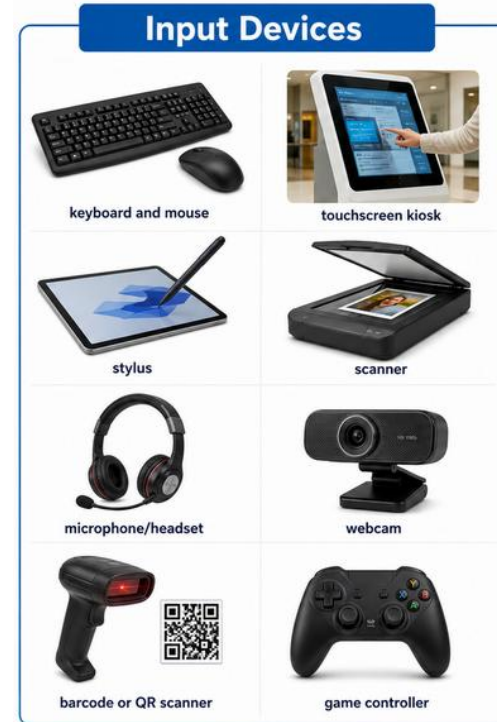


Figure 5-1 Examples of input devices

How Do You Use Output? — The Four Types of Output

Output: Output varies in form depending on the hardware and software being used and the requirements of the user. Soft copy is information on a display that exists electronically and appears for a temporary period. Hard copy (printout) is a permanent physical form. (Campbell et al., 2023)

Four basic types of output include:

- ✓ **Text:** Characters, words, and sentences displayed on screen or printed. The most fundamental and widely used output type.
 - ▶ Email messages, memos, books, reports, SMS messages, WhatsApp texts
- ✓ **Audio:** Sound output — music, speech, alerts, and audio effects produced by the computer's audio hardware.
 - ▶ Sporting events, radio broadcasts, audio clips, music streaming, voice calls, system alerts
- ✓ **Graphics:** Digital images, photos, charts, diagrams, and animated images displayed or printed.
 - ▶ Photos, clip art, charts, GIFs, infographics, maps, data visualisations
- ✓ **Video:** Moving images — sequences of frames displayed rapidly to create the illusion of motion.
 - ▶ Movies, weather forecasts, live sports performances, video calls, screen recordings

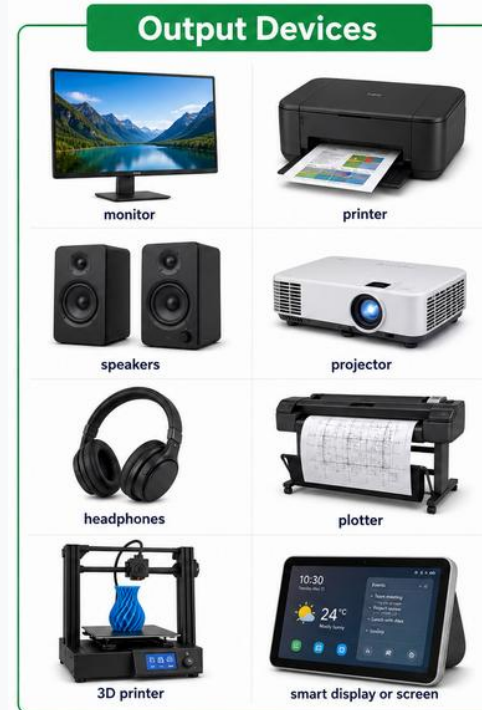


Figure 5-2 Examples of output devices

Manual Input Devices

Keyboards · Mouse · Touch screens & gestures · Pen input · Graphics tablet

Manual Input — Keyboards

Manual Input Device: One in which you enter data directly into the device. Keyboards and pointing devices are traditional types of manual input devices. Keyboards enable you to type text or commands; pointing devices enable you to make selections. (Campbell et al., 2023)

Keyboard Shortcut: A key or combination of keys used to access a feature or perform a command. Examples: Ctrl+C (copy), Ctrl+V (paste), Ctrl+Z (undo), Alt+Tab (switch window). (Campbell et al., 2023)

Additional Keyboard Features:

Many keyboards also include media control buttons (play/pause/volume), Internet control buttons (browser open/home/refresh), and other special keys: Ctrl+C/V/X/Z (copy/paste/cut/undo), Ctrl+A (select all), Alt+Tab (switch between open apps), Windows+D (show desktop).

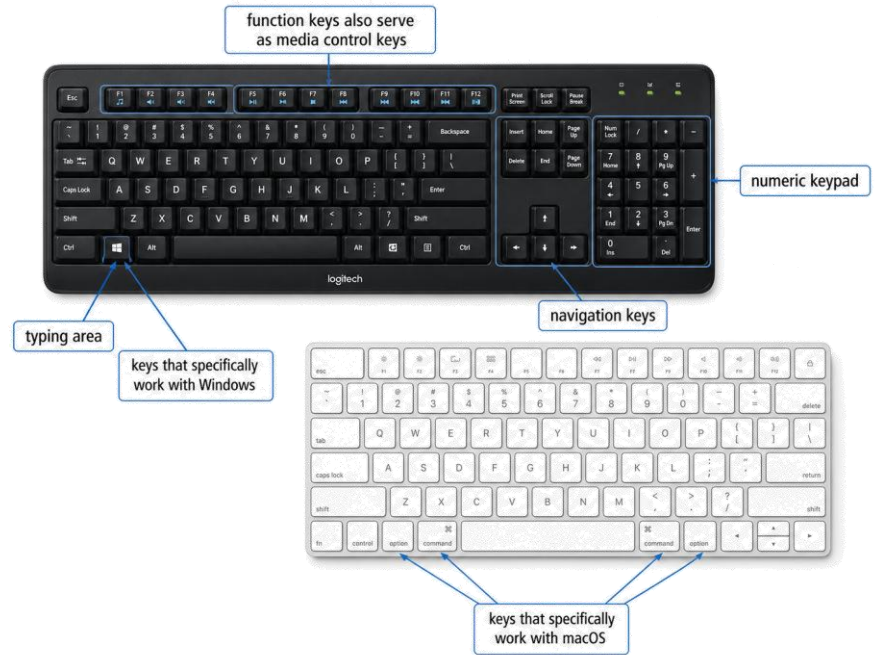


Figure 5-3 Windows and macOS standard keyboards

Manual Input — Keyboards

Manual Input Device: One in which you enter data directly into the device. Keyboards and pointing devices are traditional types of manual input devices. Keyboards enable you to type text or commands; pointing devices enable you to make selections. (Campbell et al., 2023)

All Keyboards Have These Key Areas:

Typing Area

The main alphanumeric section containing letters A-Z, digits 0-9, and punctuation. Produces visible character output when pressed. Standard QWERTY layout used worldwide.

Toggle Keys

Alternate views or turn a feature on or off each time pressed. Examples: Caps Lock (toggles uppercase) and Num Lock (toggles numeric keypad).

Function Keys

Labelled F1–F12. Special keys programmed to issue commands to a computer. Function varies by application (e.g. F5 = Refresh in browser; F2 = Rename in File Explorer).

Navigation Keys

Arrow keys, Page Up/Pg Up, Page Down/Pg Dn, Home, End. Used to move the insertion point left, right, up, or down in a program or app.

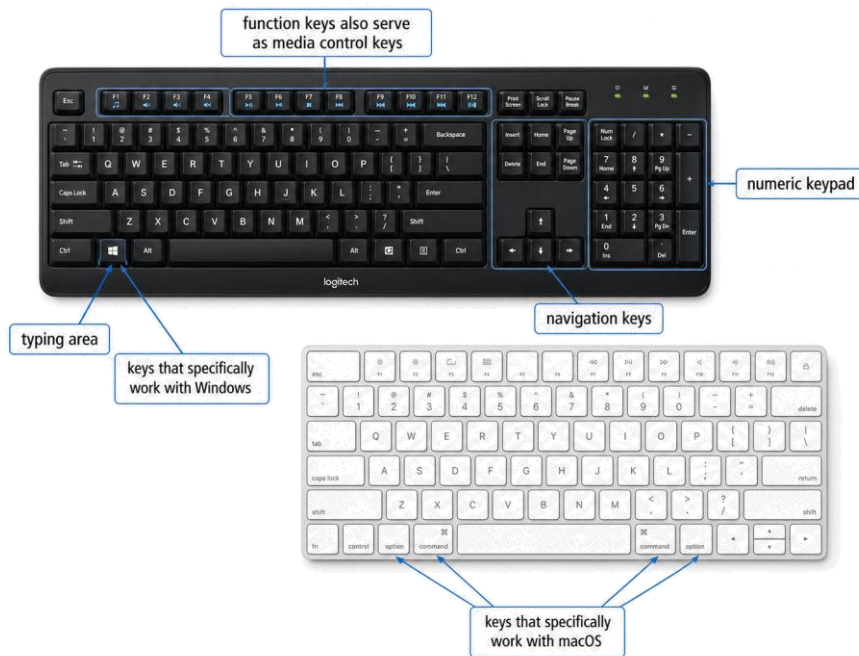


Figure 5-3 Windows and macOS standard keyboards

Manual Input — Pointing Devices

In a Graphical User Interface, a pointer is a small symbol on the screen that takes different shapes depending on the task, application, or location. A pointing device can enable you to select text, graphics, and other objects, such as buttons, icons, links, and commands. (Campbell et al., 2023)

Mouse:

A pointing device that fits under the palm comfortably. The bottom contains a mechanism that detects mouse movement.

Optical Mouse:

Uses optical sensors (LED light) to detect surface movement. Most common type. Works on most surfaces.

Laser Mouse:

Uses laser sensors for higher precision. Works on more surface types including glass.

Touch Mouse:

A touch-sensitive mouse that recognises multi-touch gestures on its surface.

Ergonomic Mouse:

Designed to reduce pain and discomfort associated with RSIs. Vertical or sculpted form reduces forearm twist.



Figure 5-4 Laser Mouse

Manual Input — Pointing Devices

Touchpad

A small, flat, rectangular pointing device that is sensitive to pressure and motion. Found most often on laptops and convertible tablets. Some touchpads also recognise touch gestures such as swipes, pinches, and stretching motions.



Figure 5-5 Touch pad

Touch Input Devices – Monitors and Screens

Touch-enabled monitors and screens allow users to interact with the operating system without a keyboard or pointing device.

- Instead of using a mouse to click on an object on the screen, users simply can tap or double-tap the item.
- Users can tap or double-tap an icon to start a program or an application.
- Slide their finger to scroll.
- Use their finger to drag items across the screen. Devices that use touch input include:
 - ✓ Monitors and screens for computers, tablets, smartphones
 - ✓ Wearable devices (smart watches — no room for physical keyboard)
 - ✓ Portable media players (maximise screen size)
 - ✓ Digital cameras (swipe to browse, tap to crop)
 - ✓ Kiosks and navigation systems
 - ✓ In-car systems (reduced functionality while moving)

Finger Input — Use a touch screen by touching areas of the screen with your finger or a stylus to make selections or to begin typing.

A gesture is a motion you make on a touch screen with the tip of one or more fingers or your hand. (Campbell et al., 2023)

Tap:



Quickly touch and release one finger once → activate link, press button, run app

Double Tap:



Two quick touches → run app; zoom in at tap location

Press & Hold:



Hold one finger → display shortcut menu; enter move mode

Drag / Slide:



Hold and move finger → move item across screen; scroll

Swipe:



Hold and move horizontally/vertically → scroll; reveal edge commands

Stretch:



Move two fingers apart → zoom in (show smaller area larger)

Pinch:



Move two fingers together → zoom out (show larger area smaller)

Manual Input —Monitors and Screens

- ✓ **Wearable devices**, such as smart watches, do not have room for a physical keyboard, so they mainly rely on touch input.
- ✓ **Portable media players** widely use touch as the primary method of input so that the size of the screen on the device is maximized.
- ✓ Touch input helps **digital camera** users perform gestures such as swiping left and right on the screen to browse photos, tapping the screen, tapping areas of photos to remove red-eye, adding a filter, and dragging borders of photos to crop them.

- ✓ **Navigation systems** allow users to perform actions such as tapping, dragging or pinching, and stretching to zoom.
- ✓ Some navigation and other in-vehicle systems **enable voice input** as well. To reduce the chances of driver distraction, some built-in navigation systems reduce functionality while the vehicle is in **motion**.



Figure 5-6 Using a touch screen on wearable device



Figure 5-7 In-car navigation system

Manual Input — Pen Input

With pen input, you touch a stylus or digital pen on a flat surface to write, draw, or make selections. Pen input devices can be used to input information on a screen. (Campbell et al., 2023)



Figure 5-8 Signing at point-of-sale terminal



Figure 5-9 Graphics tablet

Stylus

A small metal or plastic device that looks like a tiny ink pen but uses pressure instead of ink. Some stylus designs include buttons you can press to simulate clicking a mouse. (Campbell et al., 2023)

Uses:

- ▶ Touch screen navigation on tablets (iPad, Samsung Tab)
- ▶ Drawing and sketching on Microsoft Surface with Surface Pen
- ▶ Point-of-sale signature capture
- ▶ Selecting small UI elements more precisely than a finger

Digital Pen

Captures and converts a user's handwriting or drawings into a digital format. Once uploaded, handwriting recognition software translates handwritten letters and symbols into typed text or objects. (Campbell et al., 2023)

Uses:

- ▶ Medical professionals recording patient notes digitally
- ▶ Students converting handwritten lecture notes to typed text
- ▶ Architects sketching building designs with automatic digitisation
- ▶ Signature capture: software transmits signature to a central computer for storage

Graphics Tablet (Digitiser)

A hardware device used to create drawings with a pressure-sensitive pen, using an electronic plastic board that detects and converts stylus/pen movements into digital signals sent to the computer. Also called a digitiser. (Campbell et al., 2023)

Uses:

- ▶ Graphic designers creating logos, illustrations, and digital art
- ▶ Architects and engineers drawing technical diagrams
- ▶ Animators and visual artists creating frame-by-frame animation
- ▶ Photo editors retouching images with precise brush control

Digital Input Devices

Voice · Audio · Video · Motion · NFC · Scanners · OCR · Barcodes · QR · RFID · MICR

Digital Input — Voice & Audio

Voice & Audio Input

Voice Input:

The process of entering input by speaking into a microphone. The microphone may be built into the device or be an external peripheral.

Voice Recognition App:

Allows users to dictate text and enter instructions by speaking. Some mobile devices have speech-to-text that recognises spoken words and enters them into email, text messages, or apps supporting text entry.

Audio Input:

Entering any sound into a computer: speech, music, and sound effects. Requires a sound card or integrated sound capability for high-quality audio.

Music Production Software:

Allows users to record, compose, mix, and edit music and sounds. Used in professional studios and home recording setups.



Figure 5-10 Siri, Apple's voice recognition application.



Figure 5-11 Using sound mixing software to edit audio.

Digital Input — Video

Video Input

Capturing full-motion images and storing them on a storage medium or cloud. A digital video (DV) camera records video as digital signals for direct computer transfer. A webcam captures video and still images — usually includes audio input too. (Campbell et al., 2023)

6 Steps of DV Technology :

1. Select a DV camera
2. Record a video
3. Transfer and manage videos
4. Edit a video
5. Enhance a video (add special effects: warp, morphing, color change)
6. Distribute a video

Special effects include warping, changing from color to black and white, morphing, or zoom motion. Morphing transforms one video image into another image throughout several frames of video.



Figure 5-12 Using video editing software.

Digital Input — Motion

Motion Input & Gesture Recognition

With motion input, users guide on-screen elements using air gestures — moving your body or a handheld input device through the air. A device containing a camera detects the gesture and converts it to a digital signal. (Campbell et al., 2023)

Entertainment

Motion-sensing game controllers (e.g. Nintendo Wii, PlayStation Move) enable users to guide on-screen elements by moving a handheld device through the air — simulating golf swings, balance poses, sports.

Medical Field

Used for surgeon training using motion that enhances movements. Surgeons can also operate less invasively. Motion input improves precision and enables minimally invasive procedures via robotic assistance.

Military & Rehab

Military uses for flight simulation and weapon training. Also aids physical rehabilitation for wounded soldiers and assists in PTSD recovery through motion-based therapy programmes.

Sports Training

Coaches use motion input to improve athletes' performance and correct injury-causing motions. Combining motion data with complex algorithms identifies areas for improvement.

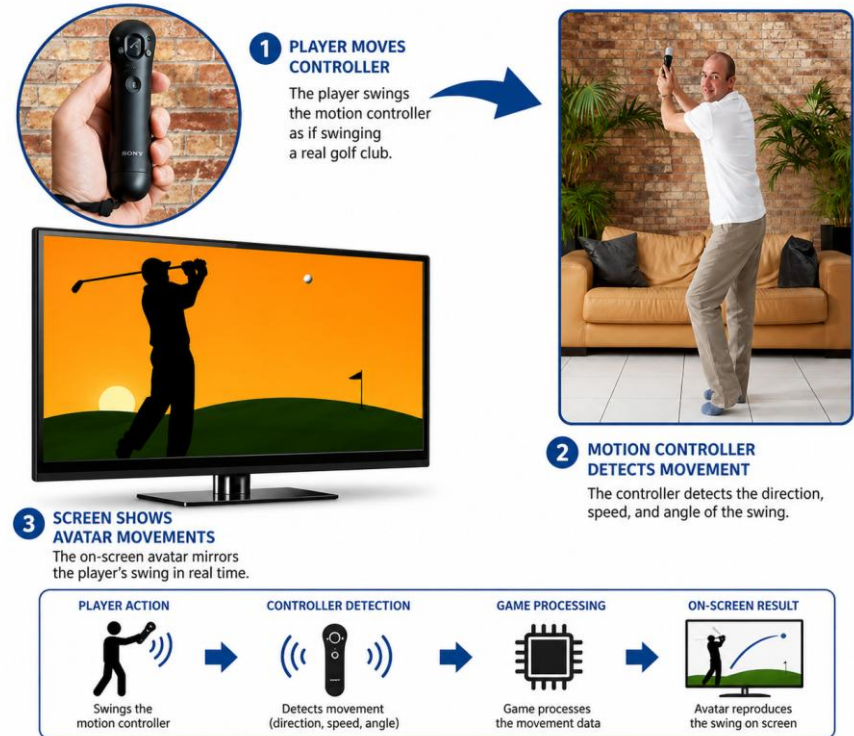


Figure 5-13 Using a motion-sensing game controller.

Digital Input — Scanners & Reading Devices

NFC— Near-Field Communication

- ✓ *NFC is a close-distance network protocol used by smartphones and other devices to read data from another device or an item, such as a credit card or ticket.*
- ✓ *NFC enables the contactless transfer of data by enabling an app to read an electronic tag.*



Figure 5-14 Using NFC to make payment.

Optical Reader

- ✓ *An optical reader is a device that uses a light source to read characters, marks, and codes and then converts them into digital data that a computer can process. Two technologies used by optical readers are:*
 - ✓ *Optical character recognition (OCR) devices read printed characters in a special font.*
 - ✓ *Optical mark recognition (OMR) devices read hand-drawn marks, such as small circles or rectangles.*

Optical Scanner

- ✓ *An optical scanner, usually called a scanner, is a light-sensing input device that reads printed text and graphics and then translates the results into a form the computer can process.*
- ✓ *The quality of a scanner is measured by its resolution, that is, the number of bits it stores in a pixel and the number of pixels per inch.*

Digital Input — Scanners & Reading Devices

Bar Code / QR — 1-D Barcode & QR Code Scanner

- ✓ A bar code reader is an optical reader that uses laser beams to read bar codes — vertical lines/spaces of different widths.
- ✓ A bar code is an identification code that often consists of either a set of vertical lines and spaces of different widths or a two-dimensional pattern of dots, squares, and other images.
- ✓ A QR code (quick response) or 2D code stores information in both vertical AND horizontal directions in a square graphic representing a web address or other content. (Campbell et al., 2023)

RFID — Radio Frequency Identification

Technology using radio signals to communicate with a tag placed on/attached to an object, animal, or person. An RFID reader reads tag information via radio waves. Readers can be handheld or mounted (e.g. in a doorway). (Campbell et al., 2023)

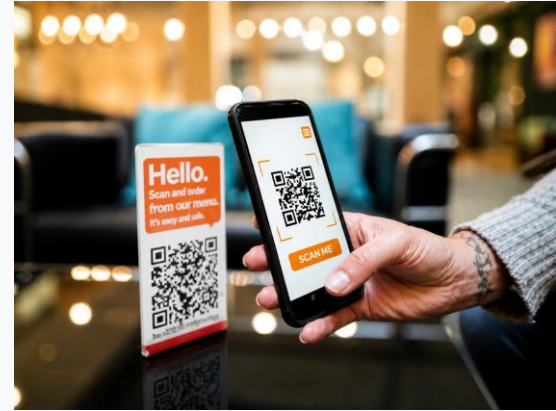


Figure 5-15 Using QR code to access information.

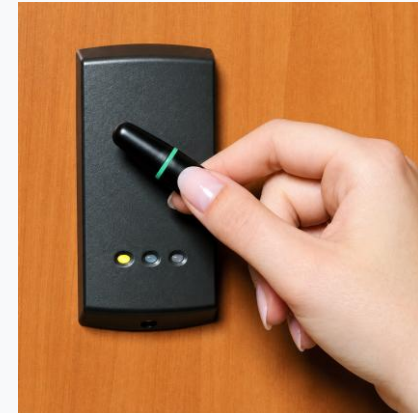


Figure 5-16 Using an RFID electronic key system.

Digital Input — Scanners & Reading Devices

Magstripe Reader

- ✓ A magstripe reader, short for a magnetic stripe card reader, reads the magnetic stripe on the back of credit cards, entertainment cards, bank cards, identification cards, and other similar cards.
- ✓ The stripe contains information identifying the owner of the card and the card issuer.
- ✓ As chip-and-pin technology becomes more prevalent, magstripe readers are used less widely.
- ✓ When a credit card is swiped through a magstripe reader, it reads the information stored on the magnetic stripe.
- ✓ **MICR** (magnetic-ink character recognition) device reads text printed with magnetized ink. MICR reader converts MICR characters into a form the computer can process.
- ✓ The banking industry almost exclusively uses MICR for check processing.
- ✓ Each check that is inserted in the MICR reader sends the check information—including the amount of the check—to a computer for processing.

Data Collection Device

- ✓ A data collection device obtains data directly at the location where the transaction or event takes place.
- ✓ These types of data collection devices are used in restaurants, grocery stores, factories, warehouses, the outdoors, or other locations where heat, humidity, and cleanliness are not easy to control.
- ✓ Data collection devices have the capability of wirelessly transmitting data over a network or the Internet.



Figure 5-17 Using a data collection device in a warehouse.

Physical Output — Printers & Hard Copy

Hard copy vs soft copy · Portrait vs landscape · Non-impact & impact printers

Physical Output — Printers & Hard Copy

Hard Copy (Printout): Printed information that exists physically — a more permanent form of output than a display (soft copy). A hard copy is either Orientation: PORTRAIT (taller, information on shorter width — letters, reports, books) or LANDSCAPE (wider, information on widest part — spreadsheets, slideshows, graphics). (Campbell et al., 2023)

Non-Impact Printers — *forms characters and graphics on a piece of paper without actually contacting the paper. Some spray ink, while others use heat or pressure to create images. Commonly used non-impact printers are:*

Inkjet Printer

Definition:

- ✓ An ink-jet printer is a type of non-impact printer that forms characters and graphics by spraying tiny drops of liquid ink onto paper.
- ✓ Inkjet printers produces text and graphics in both black-and-white and colour on a variety of paper types and sizes.
- ✓ The print head contains ink-filled cartridges with 50 to several hundred nozzles.

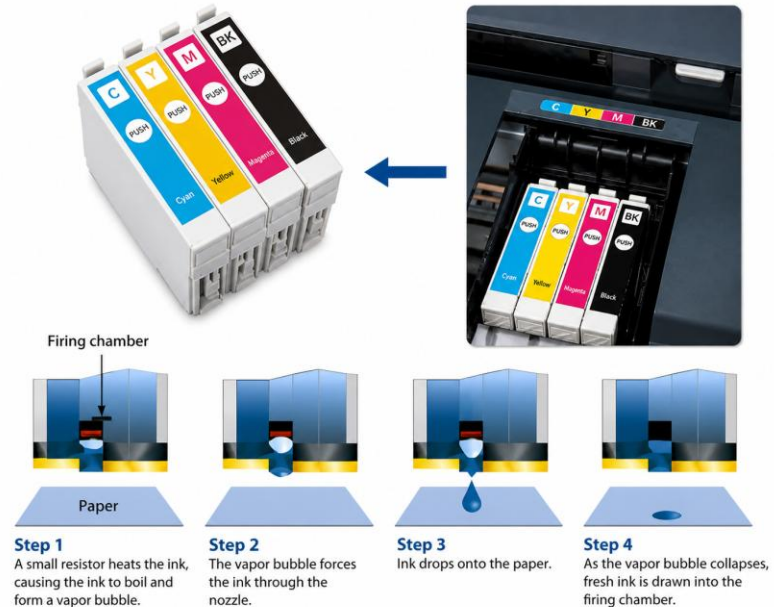


Figure 5-18 How an inkjet printer works.

Physical Output — Printers & Hard Copy

Commonly used non-impact printers are:

Photo Printer

Definition:

- ✓ A colour printer that produces lab-quality photos. Many use inkjet technology.
- ✓ Models can print letter-sized documents, allowing use for all printing needs.

Laser Printer

Definition:

- ✓ A high-speed, high-quality non-impact printer available in black-and-white and colour.
- ✓ Laser printers print text and graphics in high-quality resolutions.
- ✓ Processes and stores the ENTIRE PAGE before printing.
- ✓ Creates images using a LASER BEAM and powdered ink called TONER on a drum inside the printer.

Figure 5-19 Color printer.



Figure 5-20 How an laser printer works.

Physical Output – Printers & Hard Copy

Commonly used non-impact printers are:

All-in-One / MFP

- ✓ A single device that looks like a printer or copier but provides the functionality of a **PRINTER**, **SCANNER**, and **COPY MACHINE**. Also called a multifunction printer (MFP).

3-D Printer

- ✓ Uses **ADDITIVE MANUFACTURING** to create an object by adding material to a 3D object, one horizontal layer at a time.
- ✓ Can print solid objects: clothing, prosthetics, eyewear, implants, toys, parts, and prototypes.

Thermal Printer

- ✓ Generates images by pushing **ELECTRICALLY HEATED PINS** against heat-sensitive paper. Some have high print quality.
- ✓ A dye-sublimation printer uses heat to transfer coloured dye to specially coated paper (sometimes called a digital photo printer).

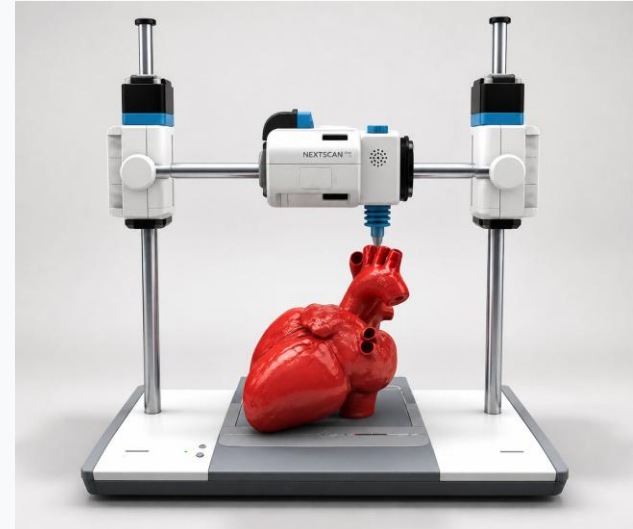


Figure 5-21 3-D printer creating a model of a heart.

Physical Output — Printers & Hard Copy

Commonly used non-impact printers are:

Mobile Printer

- ✓ *Small, lightweight, BATTERY-POWERED printer allowing a mobile user to print from a laptop, smartphone, or mobile device while travelling.*
- ✓ *Mainly uses inkjet or thermal technology.*

Plotter / Large-Format

- ✓ *A sophisticated printer producing HIGH-QUALITY DRAWINGS such as blueprints, maps, and circuit diagrams.*
- ✓ *Uses charged wires (styli) to draw electrostatic patterns.*
- ✓ *Large-format printers create photo-realistic quality prints for signs and posters.*

Other printers include:

Impact Printer

- ✓ *Forms characters by STRIKING a mechanism against an inked ribbon that physically contacts the paper.*
- ✓ *Characteristically NOISY. IDEAL for printing multi-part forms (carbon copies).*

Label Printer

- ✓ *A label printer is a small printer that prints on an adhesive-type material that can be placed on a variety of items, such as envelopes, packages, optical discs, photos, and file folders.*
- ✓ *Most label printers also print bar codes.*



Figure 5-22 A large-format printer.

Digital Output — Displays, Audio & Projectors

LCD · LED · OLED · AMOLED · Display quality metrics · GPU · Speakers · Projectors

Digital Output — Display Devices

Display Device: An output device that visually conveys text, graphics, and video information. Soft copy is information on a display that exists electronically and appears for a temporary period. Desktops use monitors; laptops and mobile devices use integrated screens. (Campbell et al., 2023)

- ✓ *Adjustable monitor stands allow you to adjust the height of the monitor to be at eye level.*
- ✓ *Many users set up multiple monitors to display separate screens of information from the same computer or device*



Figure 5-23 Using two monitors.

- ✓ *An LCD (liquid crystal display) sandwiches a liquid compound between two sheets of material that presents sharp, flicker-free images on a screen when illuminated.*
- ✓ The light source, called the backlight, often uses either CCFL (cold cathode fluorescent lamp) or LED (light-emitting diode technology).
- ✓ A display that uses an LED for the backlight is often called an LED display. LED displays consume less power, last longer, and are thinner, lighter, and brighter.
- ✓ Some displays use OLED technology which includes organic molecules that are self-illuminating and, thus, do not require a backlight.
- ✓ OLEDs can also be fabricated on thin, flexible surfaces.
- ✓ An AMOLED (active-matrix OLED) screen uses both active-matrix, that is, a high-quality, lightweight display technology capable of producing a wide range of colors with a fast response time.

Digital Output – HDTVs and Smart TVs

A Smart TV is an Internet-enabled HDTV from which you can browse the web, stream video from online media services, listen to Internet radio, communicate with others on online social media, play online games, and more.

- ✓ A plasma display uses gas plasma technology, which sandwiches a layer of gas between two glass plates.
- ✓ UHD (ultra-high-definition) television expands on HDTV technology to provide even better resolution.



Figure 5-24 Using a smart tv.

Digital Output — Display Quality Metrics

The screen on a monitor, laptop, tablet, smartphone, or other mobile device is measured diagonally from one corner to the other. When purchasing a monitor, laptop, or mobile device, these six quality metrics determine the viewing experience.

Resolution

Displays are optimised for a specific resolution called the NATIVE RESOLUTION. A higher resolution uses more pixels and provides a smoother, sharper, and clearer image. As resolution increases, some items appear smaller on screen.

Response Time

The time in MILLISECONDS (ms) that it takes to turn a pixel on or off. The LOWER the number, the faster the response time. Fast response prevents motion blur in video and games.

Brightness

Measured in NITS — a unit of visible light intensity equal to one candela per square metre (cd/m^2). Higher nits = brighter screen = better visibility in bright environments. The candela is the standard unit of luminous intensity.

Dot Pitch (Pixel Pitch)

The distance in MILLIMETRES between pixels on a display. Text created with a smaller dot pitch is easier to read. The LOWER the number, the sharper the image.

Contrast Ratio

Describes the difference in light intensity between the BRIGHTEST WHITE and DARKEST BLACK a display can produce. HIGHER contrast ratios represent colours better and produce deeper blacks.

GPU & Aspect Ratio

A graphics chip (GPU — Graphics Processing Unit) controls the manipulation and display of graphics — integrated on the motherboard or resides on a videocard in a slot on the motherboard. The ASPECT RATIO defines a display's width relative to its height. HDTV (high-definition television) works with digital broadcast signals, transmitting digital sound, supporting wide screens, and providing high resolutions.

Digital Output — Audio Output & Projection Devices

Audio Output Devices

Internal Speakers:

Most computers and mobile devices have a small internal speaker that usually emits only low-quality sound — mainly for system alerts.

External Speakers/Surround Sound:

Many users attach surround sound speaker systems to their devices to generate higher quality sound. Speaker systems connect via 3.5mm audio jack, USB, or Bluetooth.

Headphones:

Allow only the individual wearing them to hear the sound. Usually include noise-cancelling technology to block external noise. Over-ear, on-ear, and in-ear (earbuds) variants.

Earbuds:

In-ear speakers — smaller and more portable than headphones. Modern TWS (True Wireless Stereo) earbuds (AirPods, Samsung Galaxy Buds, Tecno Buds) are Bluetooth-connected with no cable.

Headset:

Functions as both headphones AND a microphone. Used for voice calls, video conferencing, gaming, and dictation. Built-in microphone enables simultaneous input + output.

Sound Card:

Integrated sound capability or separate sound card required for high-quality audio input and output. Most modern motherboards have integrated sound. Dedicated sound cards for professional audio production.



Figure 5-25 Computer speakers.

Digital Output — Audio Output & Projection Devices

Projection & Interactive Devices

Data Projector

A device that projects the text and images displayed on a computer or mobile device screen onto a LARGER SCREEN so an audience can see the image clearly. Many classrooms use data projectors so all students can see an instructor's presentation.

Interactive Whiteboard

A touch-sensitive device, resembling a dry-erase board, that displays the image on a connected computer screen, usually via a projector. Allows touching or writing on the board surface to interact with the computer display.

Gaming & Motion-Sensing Output

Joysticks, wheels, gamepads, and motion-sensing controllers are used to control the movements and actions of a player or object in computer games, simulations, and video games. They also function as OUTPUT devices when they include FORCE FEEDBACK — vibration or resistance that responds to game events.

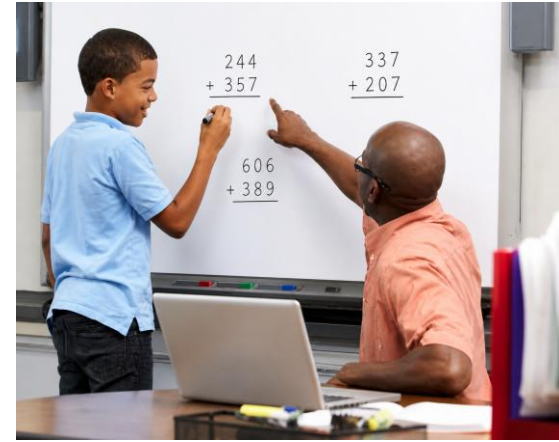


Figure 5-26 Interactive whiteboard.



Figure 5-27 Gaming wheel.

Accessibility, Customisation & E-Waste

Adaptive I/O · Braille displays · OCR readers · Device settings · Radiation & e-waste

Assistive Technology – Adaptive Input & Output Devices

The goal of accessible input and output is to enable ALL users to be independent when working with, being entertained by, or accessing services that use technology. (Campbell et al., 2023)

Adaptive INPUT Devices

Adaptive Input Devices for blind, visually impaired users, limited hand mobility, deaf users, and so on include the following:

- ✓ **Eye gaze** pointing devices analyze the area of the screen that the user is looking at.
- ✓ A **foot mouse** enables users with limited mobility in their arms or hands to control input using their feet.
- ✓ **Sip-and-puff** (SNP) technology enables users to control the mouse or other computer or device functions using a tube that they either inhale (sip) or exhale (puff) to issue commands.

Adaptive OUTPUT Devices

Adaptive Output Devices include the following:

- ✓ **Refreshable Braille displays** use nylon pins or magnets to translate content into the Braille system so that users can read it with their fingers.
- ✓ Braille printers use embossing techniques to produce hard copies of digital content that can be read by touch.
- ✓ OCR readers can scan printed materials and digitize them so that a screen reader can read the content aloud or print it on a Braille printer.



Figure 5-28 Refreshable Braille display.

Customising & Managing Input/Output Devices

*When you start using a computer or device, the operating system and related software and hardware have **DEFAULT SETTINGS** — standard settings that control how the screen is set up and how a document looks when you first start typing. Everyone customises these settings to be more productive.*

Customise System Software

- ✓ Changing the brightness of the screen
- ✓ Adding a desktop theme (wallpaper, colour scheme, icons)
- ✓ Adjusting the screen resolution
- ✓ Adding a sound scheme (system alert sounds)
- ✓ Pinning frequently used apps to the taskbar for easy access

Customise Input & Output Devices using System Software

- ✓ Controlling the pointing device with the keyboard by using the arrow and other keys
- ✓ Changing the language or dialect associated with the keyboard
- ✓ Creating new keyboard shortcuts to commands
- ✓ Adjusting the settings for toggle keys

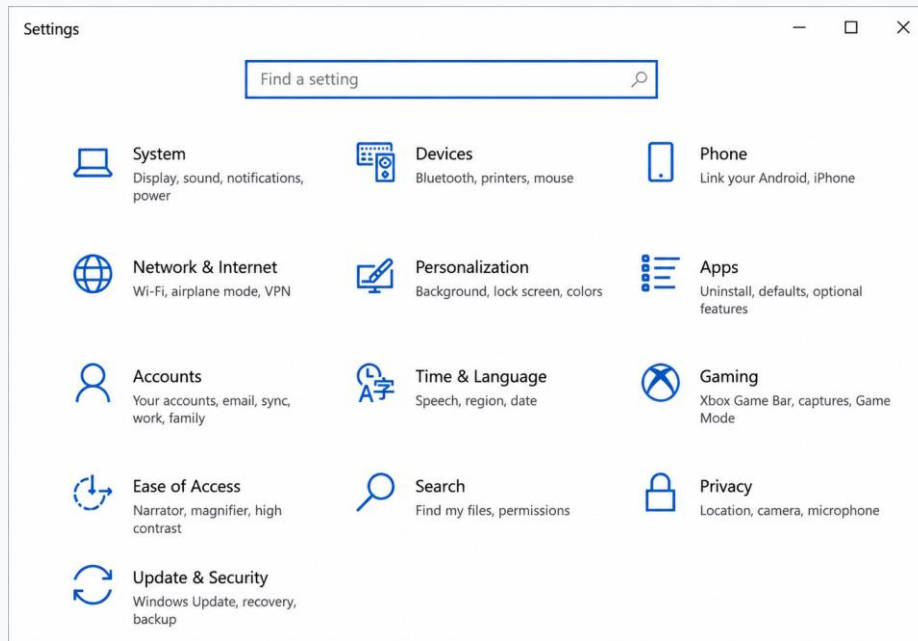


Figure 5-29 The Windows settings dialogue box.

E-Waste Risks — Radiation, Health & Responsible Disposal

Radiation from Devices: Every electronic device emits some level of radiation. The amounts for computers and mobile devices may not be harmful in low doses. However, precautionary measures reduce long-term risk and responsible disposal of old devices prevents environmental contamination.

Strategies to Protect Yourself & Minimise Risks

✓ Replace older CRT monitors with LCD monitors — older CRTs emit significantly more radiation

✓ Sit back from your monitor as far as possible (50–70 cm recommended — also good ergonomics)

✓ While using a laptop while sitting, protect your lap with a device that shields you from radiation

✓ Move other electronic sources — hard drives, printers — as far away as possible

✓ Minimise wireless connections: wireless keyboard, wireless mouse use Bluetooth/2.4GHz

✓ Remove Bluetooth headset when not in use — extended wireless device near brain is precautionary concern

✓ Turn off devices when not in use — no passive radiation from powered-off devices

✓ Recycle or donate older, unused devices — eliminate radiation exposure from old devices even when idle

✓ Take regular breaks from technology use — reduces cumulative exposure and eye strain

Tutorial Discussion Activity – Module 5

Groups of 3–4 · 15 minutes group work · Then share findings

1

A modern smartphone has MANY input and output devices. (a) List at least TEN input devices found in a typical Android smartphone sold in Zambia (e.g. Tecno Camon series). (b) List at least SIX output devices. (c) Identify TWO devices that function as BOTH input AND output simultaneously. (d) Which of your listed input devices use manual input? Which use digital input?

💡 *Think: camera, microphone, fingerprint, GPS, accelerometer, NFC, barcode scanner app, touchscreen, proximity sensor, light sensor... Vibration motor — is that input or output?*

I/O Classification

2

A Zambian NGO is purchasing 20 computers for a rural health clinic in Copperbelt Province. The clinic has: unreliable electricity (ZESCO load-shedding), a bright sun-lit room, medical staff who need to view detailed medical images, and a limited budget of K2,000 per monitor. Using ALL SIX display quality metrics, recommend and justify a specific type of monitor for this clinic.

💡 *Which display technology suits load-shedding? What brightness (nits) for a bright sun-lit room? What resolution for detailed medical images? Budget affects choices — LED LCD is likely the answer. Justify each metric.*

Display Selection

3

A Zambian government ministry is building a new e-services portal. The IT team argues 'most users are not disabled so we don't need accessibility features.' As a CS/CE student: (a) Challenge this argument using specific statistics and legal requirements. (b) Design an accessibility plan identifying at least FOUR specific adaptive input AND output features the portal should support. (c) Explain how implementing accessibility features can BENEFIT all users, not just those with disabilities.

💡 *Zambia Persons with Disabilities Act 2012. What % of Zambians have disabilities? Universal design principle — accessibility features benefit elderly users, users with temporary injuries, and mobile users too.*

Accessibility Design

4

A new Zambian startup needs to print: (1) 500 employee identity cards with photos (2) shipping labels for 1,000 packages daily (3) architectural blueprints for their new office (4) regular A4 reports and letters (5) multi-part delivery receipts requiring carbon copies. Match EACH printing need to the CORRECT printer type. Justify each choice with at least TWO technical reasons.

💡 *Identity cards with photos → which printer? Daily 1,000 labels → which specialised printer? Blueprints → A0 large format. Reports → laser. Carbon copies → which is the ONLY type that can do this?*

Printer Selection

Key Takeaways — Module 5: Input and Output

1

I/O is the human-computer interface

Every interaction with any computing device involves input (data, commands, user responses) or output (text, graphics, audio, video). Understanding I/O enables better system design.

2

Manual input comes in many forms

Keyboards (with typing area, function, toggle, navigation keys), mouse, touchpad, touch screens, stylus, digital pen, and graphics tablets each suit different tasks and users.

3

Digital input extends far beyond typing

Voice recognition, video input, motion/gesture recognition, NFC, OCR, OMR, QR codes, RFID, MICR — each captures real-world data in a format computers can process.

4

Choose the right printer for the task

Laser for volume/text. Inkjet for colour/home. Photo for quality photos. Thermal for receipts. 3-D for objects. Plotter for blueprints. ONLY impact printers produce carbon copies.

5

Display quality requires six metrics

Resolution, response time, brightness (nits), dot pitch, contrast ratio, and GPU/aspect ratio together determine screen quality. Match metrics to use case and environment.

6

Accessibility is not optional — it is professional

Adaptive I/O (eye gaze, SNP, Braille displays, screen readers) enables ALL users. Zambia's Persons with Disabilities Act and 1.3M+ citizens with disabilities make accessible design a legal and ethical requirement.

Can You Now...? — Module 5 Learning Outcomes Self-Check

Tick what you can genuinely explain without notes. Unticked = study priority for Quiz 5.

- ✓ Differentiate between input and output
- ✓ Define data, command, and user response — give an example of each
- ✓ State the four basic types of output: text, graphics, audio, and video
- ✓ Explain the difference between hard copy and soft copy
- ✓ Distinguish portrait and landscape orientation — state which suits each document type
- ✓ Identify and describe the key areas of a keyboard: typing, function, toggle, navigation keys
- ✓ Define keyboard shortcut and give three examples
- ✓ Describe four types of mouse (optical, laser, touch, ergonomic)
- ✓ Define touchpad and state where it is most commonly found
- ✓ List and describe all SEVEN touch screen gestures
- ✓ Distinguish between a stylus and a digital pen
- ✓ Define graphics tablet (digitiser) and state its primary use
- ✓ Define voice input and voice recognition app
- ✓ Explain digital video input and list the 6 steps of DV technology

- ✓ Define motion input and name THREE sectors that use it
- ✓ Define NFC and explain how it enables contactless payments
- ✓ Distinguish OCR from OMR — give a Zambian example of each
- ✓ Define QR code and explain how it differs from a 1-D barcode
- ✓ Define RFID and MICR — state the primary industry for each
- ✓ Name at least EIGHT types of non-impact printers
- ✓ Explain what additive manufacturing is and how 3-D printing works
- ✓ Define LCD, LED, OLED, and AMOLED display technologies
- ✓ Define and apply all SIX display quality metrics (resolution, response time, brightness, dot pitch, contrast ratio, GPU/aspect ratio)
- ✓ Describe data projector and interactive whiteboard as output devices
- ✓ Name three adaptive INPUT devices and explain how they work
- ✓ Describe refreshable Braille display and Braille printer as adaptive output
- ✓ List at least SIX strategies to reduce radiation exposure from devices
- ✓ Explain the three SEM steps with Zambian examples

Practice Questions – Module 5

Prepare these for Quiz 5 and end-of-semester examination.

Input Fundamentals

- (a) Define data, command, and user response. Give ONE example of each from a Zambian digital service.
- (b) Distinguish manual input from digital input. Give THREE examples of each type.
- (c) List ALL SEVEN touch screen gestures with description and common use.
- (d) Define stylus, digital pen, and graphics tablet. How does each differ from the others?

Scanning & Reading Devices

- (a) Distinguish OCR from OMR. Give a specific Zambian educational example of OMR in use.
- (b) Define QR code and explain how it differs from a 1-D barcode. Give TWO Zambian examples.
- (c) Define RFID and NFC. How do they differ? Give one Zambian example of each.
- (d) What is MICR? Which industry uses it and for what specific purpose?

Printers & Output

- (a) A Zambian company needs to print 10,000 patient records daily. Which printer type is most appropriate? Justify using TWO technical criteria.
- (b) What is additive manufacturing? How does a 3-D printer work? Name TWO uses in Zambia.
- (c) Define thermal printer. Why does every Zambian supermarket use thermal printing for receipts?
- (d) ONLY one printer type can produce carbon copy (multi-part) forms. Which type? Why can no other type do this?

Displays & Accessibility

- (a) Distinguish LCD, LED, OLED, and AMOLED. What is the key advantage of OLED over LED?
- (b) A display has: resolution 1280×720, brightness 150 nits, response time 16ms, contrast ratio 500:1. Identify TWO quality weaknesses and suggest better specifications.
- (c) Define refreshable Braille display and Braille printer. Who uses each and how?
- (d) List FOUR strategies to reduce radiation exposure from computing devices.